MIL-C-23157D
INT. AMENDMENT-3(AS)
30 January 1985
SUPERSEDING
MIL-C-23157D
INT. AMENDMENT-2(AS)
26 January 1983

MILITARY SPECIFICATION

INTERIM AMENDMENT

CLIPBOARD, PILOT'S, MXU-163/P AND MARK 2A

This Interim Amendment is issued for use by the Department of the Navy with Military Specification MIL-C-23157D dated 23 July 1979.

PAGE 2

2.1 Issues of documents. Under "SPECIFICATIONS FEDERAL", add "L-P-1183 Plastic Molding Material, Acrylonotrile Butadiene Styrene (ABS) Rigid.

PAGE 3

Delete "MIL-T-5422 Testing, Environmental, Airborne Electronic and Associated Equipment" and "MIL-F-21840 Fastener Tapes, Hook and Pile, Synthetic." Add "MIL-P-46181 Plastic Molding Material, Polyamide (Supertough Nylon)."

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2.2 Other publications. Add at the end of the paragraph:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| ASTM B 117-73 | Salt Spray (Fog) Testing |
| ASTM D 2249-68 | Testing Coated Metal Specimens at 100 Percent Relative Humidity |
| ASTM D 4066-82 | Materials, Nylon Injection & Extrusion |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

PAGE 5

3.3.3.2 Alluminum alloy parts. After "MIL-C-5541," add "or finished with a lusterless black finish as specified in 3.4.3.5."

FSC 8465



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3.4 Design and construction.

At the end of the first sentence, add "except for the buckle."
At the end of the second sentence, add "except for the buckle."
At the end of the fifth sentence, add "or, as an alternate, two pencil holders, one on each side, will be permitted."

3.4.1 Clipboard base. Delete and substitute:

"3.4.1 Clipboard base. A clipboard base conforming to Figure 2 shall be provided for both lighted and unlighted clipboards. The base shall be of ST Nylon conforming to MIL-P-46181 to be molded a minimum of .080 inch thick in lusterless black color no. 37038 of FED-STD-595 or shall be of aluminum sheet .032 inch thick conforming to QQ-A-250/8, temper 0. The underside shall be formed with a radius of 3-1/2 inches to fit the pilot's thigh except that a slight protrusion will be permitted to accommodate the configuration of the light switch of the lighted clipboard. The inside dimensions of the base shall be 8-7/8 ±1/32 inches long by 4-7/8 ±1/32 inches wide by 1-1/8 ±1/32 inches deep."



3.4.1.4 Pilot identification-lighted clipboard. Delete and substitute:

"3.4.1.4 Pilot identification and battery insertion. A nameplate with the printed words "PILOT IDENTIFICATION" and "ALL BATTERIES MUST BE LOADED IN THE SAME DIRECTION" shall be attached to the left side of the lighted clipboard as shown in Figure 1. The nameplate shall be 2 1/4 inches long, 5/8 inch wide and 0.010 inch thick of white vinyl plastic conforming to L-P-535, Composition A, Type I, Class 2 or Type II, Grade GU with an adhesive backing."

3.4.2 Clipboard cover. Delete and substitute:

"3.4.2 Clipboard cover. The cover for the lighted clipboard shall conform to Figure 1. The cover for the unlighted clipboard shall conform to Figure 3. The cover for both clipboards shall be of ST Nylon conforming to MIL-P-46181 to be molded a minimum of .080 inch thick in lusterless black color no. 37038 of FED-STD-595 or shall be of aluminum sheet .032 inch thick conforming to QQ-A-250/8, temper 0. The cover shall be 9 ±1/32 inches long by 5 ±1/32 inches wide with a minimum of 1/4 inch formed edge. The cover shall be attached to the clipboard base with a minimum of two 1/8 inch blind rivets conforming to MIL-R-24243/1B or 3B, or with a shear fit utilizing cutouts and indentations for a snap-in assembly, or with a cut-out snap fit utilizing four plastic snaps conforming to ASTM D 4066-82. Shear fit cutouts & indentations, rivets, or plastic snaps shall be located as shown in Figure 2."



3.4.2.7 Pencil tube and pencil. Add after the first sentence, "A pencil tube may also be provided on the left side of the clipboard in addition to the one on the right side."



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3.4.3 Light assembly. Delete and substitute:

"3.4.3 Light assembly. A battery-powered, rotary switch-operated light assembly shall be attached to the upper clip of the lighted clipboard. When lighted, the light assembly shall illuminate the entire area of the clipboard cover between the upper and lower clip with white light with minimal light diffusion on either side and no upward glare. The light assembly shall contain a fresnel lens area, and shall be as illustrated in Figure 11. The thickness of the side wall and top shall be a minimum of .100 inch. The bottom of the light assembly shall be completely enclosed for maximum light reflection and to prevent stray bottom light. The complete light assembly shall be securely attached to the .040 inch thick upper clip from the bottom to brass threaded inserts that are fit into two blind holes which may be drilled and/or molded into the plastic diffuser or, as an alternate, attached with two 3/8-inch long high riser blunt end screws with a hex head and a washer attached to the head. The major diameter of the screw shall be .119 to .125. The light sockets shall be attached to an aluminum back plate which conforms to Figure 12. The back plate shall be securely attached to the rear of the light diffuser into two brass threaded inserts that are fit into two blind holes which may be drilled and/or molded into the rear of the plastic diffuser. The aluminum back plate shall be fitted with a coil spring pencil holder as outlined in 3.4.2.8. The plastic light diffuser shall be made of acrylic plastic conforming to L-P-380, except that it shall be translucent to minimize glare. The remainder of the light assembly shall be acrylic plastic conforming to L-P-380 with a lusterless black finish or a black high impact plastic conforming to L-P-1183. Aluminum parts shall conform to QQ-A-250/8 and brass threaded inserts shall be standard commercial grade. The light assembly shall be annealed at 194°F for a minimum of 4 hours after all the brass inserts have been embedded into the plastic. Any adhesive or other bonding method used in the light assembly shall be capable of withstanding all the environmental tests of this specification."

3.4.3.4 Lighting circuit. Line 6, delete "Leecraft Model 10-08," and substitute "Leecraft 500 Series."

3.4.4 Finish. Delete and substitute:

"3.4.4.5 Finish. Aluminum parts of the clipboard shall be finished with a lusterless black material, color No. 37038 of FED-STD-595. The finish for the acrylic parts shall be an acrylic sign paint which is compatible with acrylic plastics. The finish shall withstand the salt spray and humidity tests of 4.5.4."



3.4.5.1 Adjustable leg strap. Line 6, delete "tongue" and substitute "male portion of the buckle which fits into the female portion of the buckle."

Line 8 and line 12, delete "boxstitch" and substitute "cross boxstitch."



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3.4.5.2 Non-adjustable leg strap. Line 8 and line 9, delete "boxstitch" and substitute "cross box stitch."

3.4.5.3 Hook fastener tape. Delete.

3.4.5.4 Pile fastener tape. Delete.

3.4.5.5 Fastener tape. Delete.

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3.4.5.6 Stitching. Line 3, delete "boxstitch" and substitute "cross box stitch."

3.4.5.7 Buckle. Delete and substitute:

"3.4.5.7 Buckle. The leg strap buckle shall be of the quick release type and shall conform to MIL-S-43835 and Drawing 4-1-563 except that the color shall be black. The buckle shall withstand a 100-pound load for at least 10 seconds without breaking or coming unfastened by application of uniformly increasing tension as described in 4.5.1."

3.7 Instruction diagram. Delete.

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4.4.2.1.1 Examination of the end item for defects in material, construction, appearance, and workmanship. Add the following:

| Examine | Defect | Major | Minor |
|---|---|---|---|
| Construction | Batteries cannot be gravity fed or gravity removed from the battery holder tube. | X | |

PAGES 18 and 19

4.5.4 Salt spray and humidity tests. Delete and substitute:

"4.5.4 Salt spray and humidity tests.

4.5.4.1 Salt spray. One complete clipboard shall be subjected to the salt spray test as specified in ASTM B 117-73 for a period of 48 hours.

4.5.4.2 Humidity. Another complete clipboard shall be subjected to the humidity tests as specified in ASTM D2247-68. The internal temperature of the test chamber shall be at standard ambient with uncontrolled humidity. Gradually raise the internal chamber temperature to 71°C (160°) and the relative humidity to 95 percent over a period of two hours. The conditions shall be maintained not less than six hours. Maintaining 85 percent, or greater, relative humidity, reduce the internal chamber temperature in 16 hours to 28 ± 10°C (82 ± 18°F). This test procedure shall be repeated for 10 continuous cycles (not less than 240 hours).



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4.5.4.3 Examination and test. Within one hour after completion of the above tests, the lighted clipboards shall be tested for operation of the lighting circuit. Both units shall be examined for deterioration of the protective coating and finishing material applied to the aluminum surfaces. Any cracking or peeling of the protective coating or finishing material shall be scored as a defect. These environmental tests are not to be regarded as a test of any other characteristic of the clipboard."

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6.3 Data. Line 8, delete "NavWeps Form 4200/15" and substitute "NAVAIR Form 4200/30."

PAGE 23

Delete Figure 2 and substitute new Figure 2.

PAGE 27

Delete Figure 6.

PAGE 28

Delete Figure 7.

PAGE 29

Delete Figure 8.

PAGE 32

Add new Figure 11 and Figure 12.

Preparing Activity
Navy - AS
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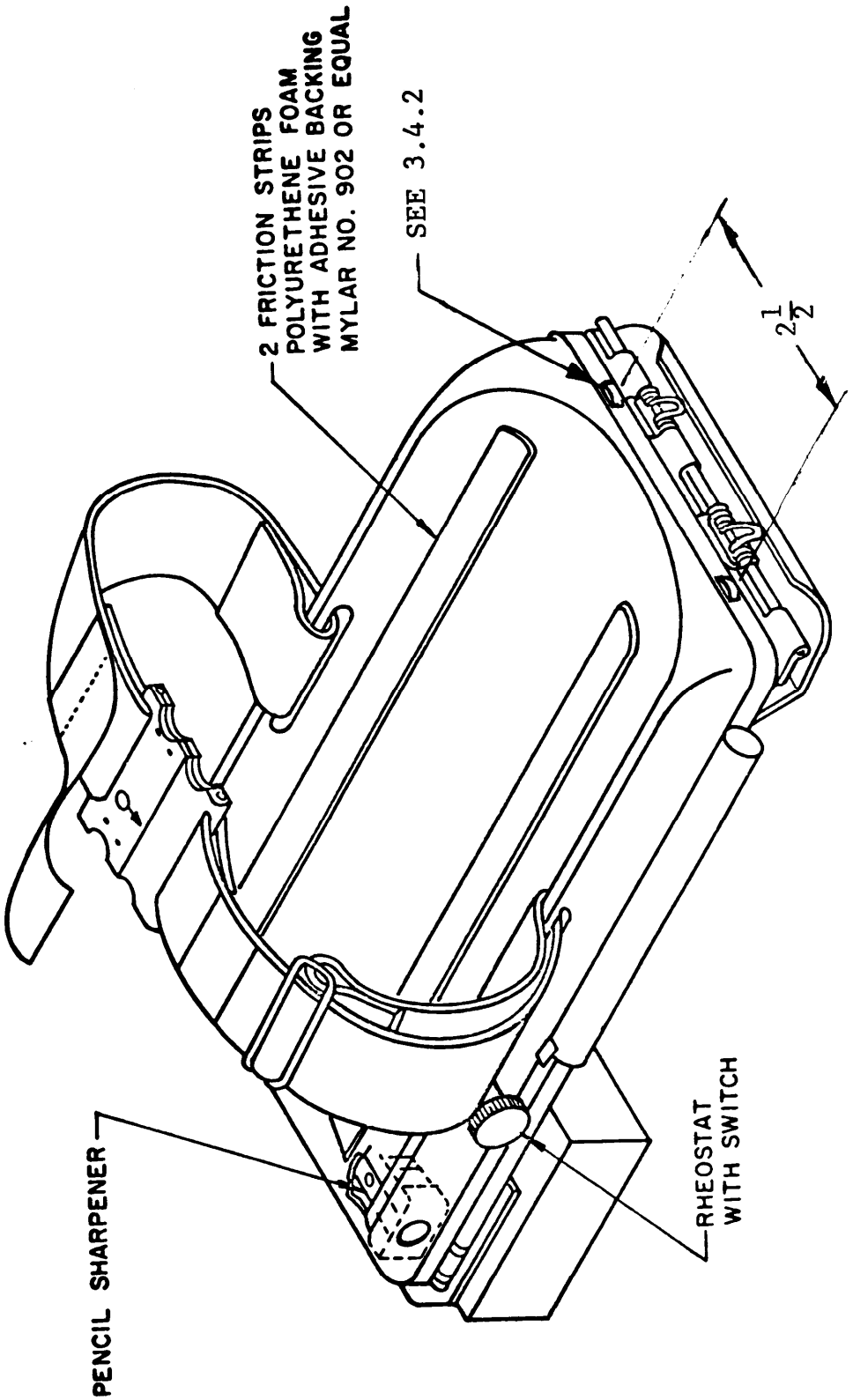



Figure 2. Pilots clipboard (MXU-163/P and MARK 2A) under side.



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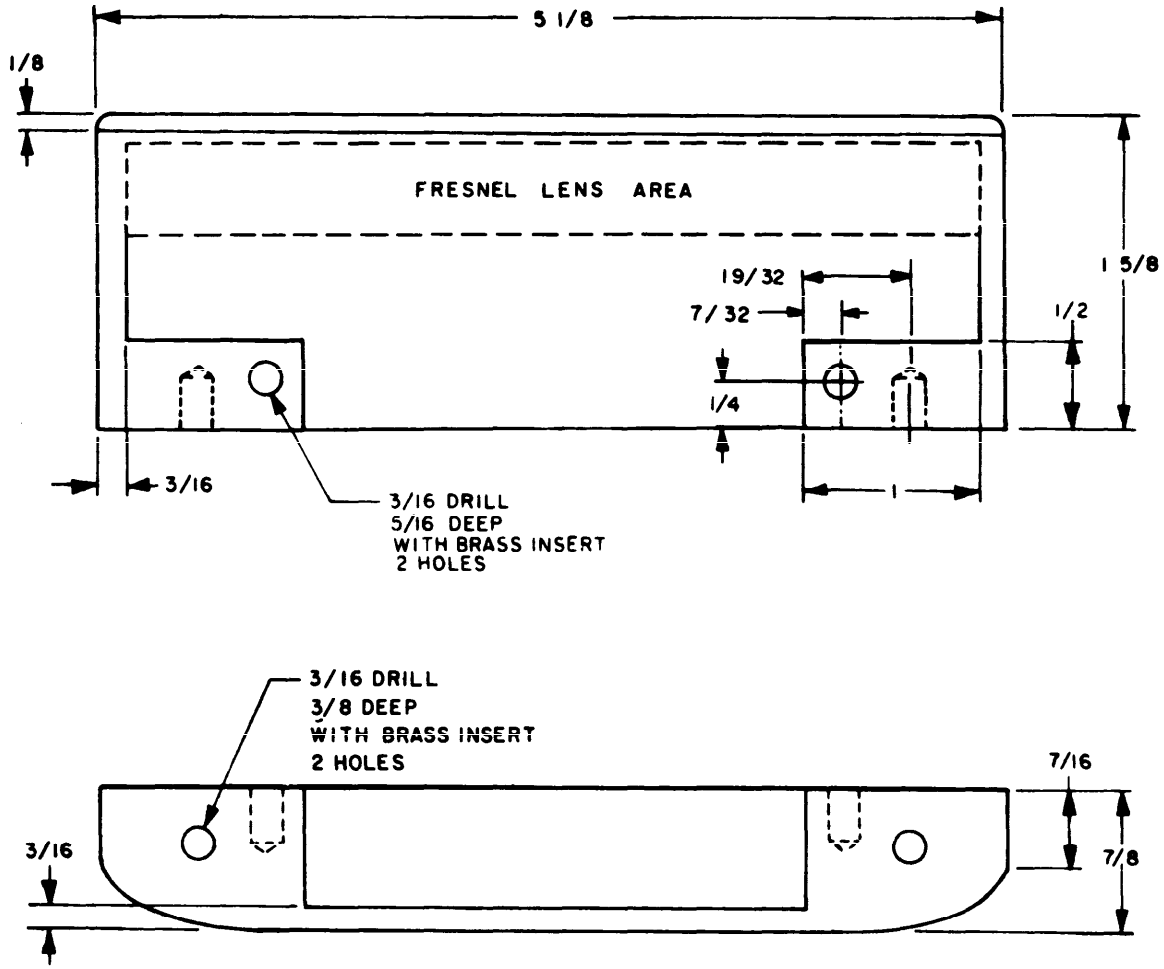



FIGURE 11.

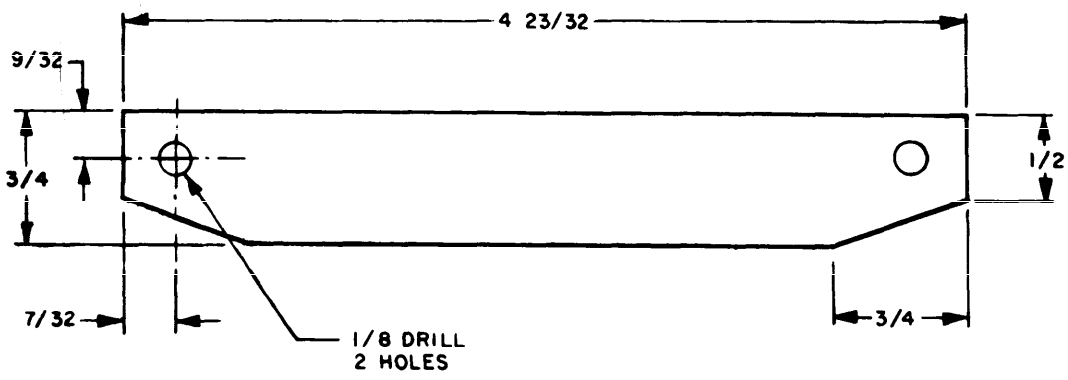



FIGURE 12.



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